Clock-Scale Ambiguity, Static CAR States, and Gibbs Modular Equivalence

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Formal statements

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Abstract. A time-translation representation determines evolution with respect to a chosen parameter but does not identify an externally calibrated physical time unit. In a bounded two-band fermionic model, multiplying a Bloch Hamiltonian by a strictly positive measurable scalar preserves its zero-temperature occupied projector, while specified dynamical or thermal observations can distinguish the resulting models. We prove a calibrated-parameter derivative identity and a finite-dimensional Gibbs/modular equivalence theorem with explicit sign conventions. These are exact mathematical statements, not claims that an experimental calibration or physical time standard has been implemented.

Keywords: time-translation parameter; CAR quasifree state; band projector; Gibbs state; modular automorphism; identifiability.

1. Assumptions and conventions

Let $\mathcal{H} = \ell^2(\mathbb{Z}; \mathbb{C}^2)$, with Fourier variable $k \in [-\pi, \pi]$. For $m \geq 0$ define the bounded self-adjoint Bloch multiplier

$$h_m(k) = \sin k\, \sigma_x + (m + 1 - \cos k)\sigma_z, \qquad E_m(k) = \sqrt{m^2 + 4(m+1)\sin^2(k/2)}.$$

The negative-band covariance is $P_m = \mathbf{1}_{(-\infty,0)}(H_m)$, using the *strictly negative* spectral convention. At $m = 0$, the sole zero is at $k = 0$, a set of measure zero for the full-chain Fourier multiplier. Gauge-invariant quasifree states are taken on one fixed CAR algebra, and measurements of varying Hamiltonians are not identified with measurements of one fixed observable [1].

2. Continuous additive changes of the time parameter

Proposition 1 (positive linear rescaling). Let $t : \mathbb{R} \to \mathbb{R}$ be continuous, additive, nonconstant and orientation-preserving. Then there is a unique $\tau_0 > 0$ such that $t(s) = \tau_0 s$. If G is self-adjoint, $U(s) = e^{-isG}$, and one chooses the physical-time convention $U(s) = e^{-it(s)H_{\mathrm{phys}}/\hbar}$, then one possible generator convention is

$$H_{\mathrm{phys}} = \frac{\hbar}{\tau_0} G.$$

Any additive scalar shift of the Hamiltonian leaves adjoint automorphisms unchanged. Uniqueness *up to* an additive scalar requires sufficiently complete observable access, for example the full algebra $B(\mathcal{H})$ or an irreducibly represented observable algebra. On a proper observable subalgebra, further generator ambiguities may arise. The translation group law does not determine τ_0.

Proof. Additivity gives $t(q) = qt(1)$ for every rational q, and continuity extends this equality to every real s. Orientation preservation and nonconstancy imply $t(1) > 0$. Substitution establishes the asserted unitary realization. Multiplying the unitary family by a scalar phase does not alter its adjoint action on observables. For a proper algebra, consider $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and $\mathcal{A} = B(\mathbb{C}^2) \otimes I$. The generators $G_1 = \sigma_z \otimes I$ and $G_2 = \sigma_z \otimes I + I \otimes \sigma_z$ induce identical adjoint automorphisms of every $A \in \mathcal{A}$, although $G_2 - G_1$ is not scalar. This explicitly shows why the observable algebra must be specified when claiming generator uniqueness. □

This is a statement about admissible parameter identifications, not a measured definition of a clock. Metrological calibration additionally concerns indications, measurement standards and associated uncertainties [2].

3. Exact invariance of zero-temperature state data

Proposition 2 (positive scalar multipliers). Suppose $a(k)$ is bounded, real, measurable and strictly positive almost everywhere. Let $\widetilde{H}_m$ be the Fourier multiplier with symbol $a(k)h_m(k)$. Then

$$\mathbf{1}_{(-\infty,0)}(\widetilde{H}_m) = P_m.$$

Consequently the corresponding gauge-invariant quasifree states, all their finite-interval restrictions and all expectations of *fixed* CAR observables coincide.

Proof. Almost everywhere, $h_m(k)$ and $a(k)h_m(k)$ have the same eigenvectors and the same eigenvalue signs. Their negative spectral projections are therefore identical almost everywhere, proving equality as decomposable bounded operators. Equality of the one-particle covariance fixes all gauge-invariant quasifree correlations by the CAR/Wick prescription [1]. Restriction to any fixed subalgebra preserves equality. □

The multiplier may vary with momentum and therefore need not represent a change of time units. The proposition does not assert equality of finite **open-chain** ground-state projectors obtained by truncating the Hamiltonians, nor equality of thermal states at fixed inverse temperature. At a finite-dimensional zero eigenspace, any separate convention for occupation of zero modes must be kept fixed. Negative scaling has no analogous projector-invariance property.

For example, $a(k) = 1 + \frac{3}{5}\cos k$ is bounded below by $\frac{2}{5}$ and changes the one-particle dispersion while keeping the occupied projector. This yields a finite-range deformation, not a physical-scale identification.

4. An ideal calibrated two-time observable

Fix a positive rate γ and define the annihilation-operator Heisenberg convention so that the associated one-particle evolution is $e^{-i\gamma H_m t}$. In the quasifree ground state, write the on-site vacancy correlation matrix as

$$F_\gamma(t;0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-i\gamma h_m(k)t}(I - P_m(k))\,dk,$$

and set $f_\gamma(t) = \mathrm{tr}_{\mathbb{C}^2} F_\gamma(t;0)$.

Proposition 3 (ideal generator-rate identification). For each fixed finite $m \geq 0$,

$$f_\gamma(0) = 1, \quad f'_\gamma(0) = -i\gamma\overline{E}_m, \quad \overline{E}_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} E_m(k)\,dk > 0.$$

Hence, for the specifically defined ideal observable and a calibrated time variable, $\gamma = if'_\gamma(0)/\overline{E}_m$. At $m = 0$, $\overline{E}_0 = 4/\pi$.

Proof. Except at the measure-zero band crossing when $m = 0$, $I - P_m(k)$ is the rank-one positive-energy spectral projection. Therefore $f_\gamma(t) = (2\pi)^{-1}\int e^{-i\gamma E_m(k)t}dk$. Since E_m is bounded, differentiation under the integral is legitimate and gives the first two equalities. For $m = 0$, $E_0(k) = 2|\sin(k/2)|$, whose average equals $4/\pi$. □

This identity presupposes the specified preparation, observable and calibrated physical-time argument. The mathematics does not show that a device can prepare the state or measure this complex correlation with the necessary precision. If m is unknown, the derivative identifies only the product $\gamma\overline{E}_m$ until another observation determines m.

5. Finite Gibbs states and the sign of modular evolution

Proposition 4 (finite Gibbs equivalence). Let H, H' be finite-dimensional self-adjoint many-body Hamiltonians and $\beta, \beta' > 0$. Define the faithful normalized density matrices

$$\rho = Z^{-1} e^{-\beta H}, \qquad \rho' = (Z')^{-1} e^{-\beta' H'}.$$

With $\sigma_s(A) = \rho^{is} A \rho^{-is}$ and $\alpha_t(A) = e^{itH/\hbar} A e^{-itH/\hbar}$, one has

$$\sigma_s = \alpha_{-\hbar\beta s}.$$

Moreover $\rho = \rho'$ if and only if $\beta H - \beta' H' = cI$ for some real scalar c.

Proof. The scalar normalization Z cancels from $\rho^{is} A \rho^{-is}$, leaving $e^{-i\beta s H} A e^{i\beta s H}$. The normalized faithful Gibbs matrices are equal exactly when their principal self-adjoint logarithms are equal. Thus $-\beta H - (\log Z) I = -\beta' H' - (\log Z') I$, which is equivalent to the stated scalar-identity relation. Conversely that relation makes the normalized exponentials equal. □

For a finite *number-conserving free-fermion grand-canonical Gibbs state at fixed chemical-potential convention*, the one-particle covariance is $C = (I + e^{\beta h})^{-1}$; when $0 < C < I$, its logit determines the dimensionless one-particle generator. This does not identify a separately calibrated physical Hamiltonian, and it is not a formula for an arbitrary interacting Gibbs state or an arbitrary compressed infinite-volume thermal covariance. Adding a scalar to the full many-body H differs from adding a scalar to one-particle h, which shifts the many-body number operator under second quantization.

6. Scope of the results

The four propositions establish exact equivalences or inversion identities under explicitly declared assumptions. They do not identify a physical clock, establish a new model of time, or promote conditional infinite-volume estimates into theorems. In particular, the equality of static projectors is logically independent of the estimates needed to compare distinct-mass relative entropies.

References

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